

Chapter 1

TT

1.1 CIC System

We first introduce some symbols in order to construct a formal language:

- ◇ $V = \{v_0, v_1, \dots\}$ is the set of term variables;
- ◇ $A = \{\alpha_0, \alpha_1, \dots\}$ is the set of type variables (below we take the union of A and V , and denote it by V);
- ◇ $C = \{\text{all UTF-8 strings}\}$ is the set of constants that are new definitions;
- ◇ $I = \{\text{Ind, Cstr, Elim}\}$ is the set of symbols used in defining inductions;
- ◇ $S = \{\text{Set, Prop}\} \cup \{\text{Type}_i\}_{i \geq 0}$ is the set of sorts;
- ◇ $P = \{., :, \equiv, (,)\}$ is the set of punctuation;
- ◇ $B = \{\lambda, \Pi\}$ is the set of binders.

Then we let

$$F = C|V|S|BV : F.F|ICF|FF$$

Axioms $\mathcal{A} \subset S^2$ and rules $\mathcal{R} \subset S^3$ are set as follows

$$\begin{aligned} \mathcal{A} &= \{(\text{Set}, \text{Type}_0), (\text{Prop}, \text{Type}_0), (\text{Type}_i, \text{Type}_{i+1})\} \\ \mathcal{R} &= \{(\text{Prop}, \text{Prop}, \text{Prop}), (\text{Set}, \text{Prop}, \text{Prop}), (\text{Type}_i, \text{Prop}, \text{Prop}), \\ &\quad (\text{Prop}, \text{Set}, \text{Set}), (\text{Set}, \text{Set}, \text{Set}), (\text{Type}_i, \text{Type}_j, \text{Type}_{\max\{i,j\}})\}. \end{aligned}$$

Note that $\Pi(\mathbf{x} : \mathbf{A})\varphi$ is the abbreviation of $\Pi x_0 : A_0. \Pi x_1 : A_1 \dots \Pi x_m : A_m. \varphi$. $\mathbf{x} : \mathbf{A}$ is the abbreviation of $x_0 : A_0 x_1 : A_1 \dots x_m : A_m$.

Definition 1.1.1 *The Arity has the form*

$$A = S|\Pi(V : F).A.$$

For any $X \in C$, the Strictly Positive Types has the form

$$P_X = X|P_X F(X \notin F)|\Pi V : F. P_X(X \notin F).$$

The Constructors has the form

$$C_X = X|C_X F(X \notin F)|\Pi V : F. C_X(X \notin F)|\Pi V : F. C_X(V \notin C_X).$$

Any $h \in C_X \setminus P_X$ is called recursive.

Definition 1.1.2 A definition D has the form

$$\mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a$$

note that $d(\mathbf{x})$ refers $d \in \mathbf{F}$ and $\text{Fr } d \subset \mathbf{x}$, that $x_i \in V, c \in C$ and that all A_i, d, a are in $\mathbf{F}$.

A environment Δ has the form $D_0 D_1 \cdots D_n$.

Definition 1.1.3 A judgment has the form

$$\Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : a$$

Definition 1.1.4 Given $\Delta = D_0 D_1 \cdots D_n$, $R_\Delta \subset \mathbf{F}^2$ is the smallest relation satisfies the following properties:

- ◇ contains $\{(c\mathbf{U}, d(\mathbf{U})) : \mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a \in D \in \Delta \text{ and } \mathbf{U} \in \mathbf{F}^*\}$;
- ◇ closed under the following operations: $\mathbf{F}-, -\mathbf{F}, \lambda V : \mathbf{F}. -, C\mathbf{p}_1 - \mathbf{p}_2$.

Definition 1.1.5 R_β is the smallest relation satisfies

- ◇ contains $\{((\lambda x : a.\varphi)\chi, \varphi_x^\chi) : x \in V \text{ and all } \varphi, \chi, a \in \mathbf{F}\}$;
- ◇ closed under the following operations: $\mathbf{F}-, -\mathbf{F}, BV : \mathbf{F}. -, BV : -.\mathbf{F}, C\mathbf{p}_1 - \mathbf{p}_2$.

In COC, we have introduced the following sequent rules.

(Sort)

If $(s_1, s_2) \in \mathcal{A}$, then

$$\frac{}{} \quad \frac{}{} \quad s_1; s_2$$

(Var)

If $x \in V \setminus \mathbf{x}$, then

$$\frac{\Delta \quad \Gamma \quad a : s}{\Delta \quad \Gamma \quad x : a \quad x : a}$$

(Weak)

If $x \in V \setminus \mathbf{x}$, then

$$\frac{\Delta \quad \Gamma \quad \varphi : a \quad \Delta \quad \Gamma \quad b : s}{\Delta \quad \Gamma \quad x : b \quad \varphi : a}$$

(Form)

If $(s_1, s_2, s_3) \in \mathcal{R}$, then

$$\frac{\Delta \quad \Gamma \quad a_1 : s_1 \quad \Delta \quad \Gamma \quad x : a_1 \quad a_2 : s_2}{\Delta \quad \Gamma \quad \Pi x : a_1. a_2 : s_3}$$

(Appl)

$$\frac{\Delta \quad \Gamma \quad \varphi : \Pi x : a. b \quad \Delta \quad \Gamma \quad \chi : a}{\Delta \quad \Gamma \quad \varphi\chi : b_x^\chi}$$

(Abst)

$$\begin{array}{ll}
\Delta \quad \Gamma \quad x : a & \varphi : b \\
\Delta \quad \Gamma \quad \Pi x : a.b : s & \\
\Delta \quad \Gamma \quad (\lambda x : a.\varphi) : (\Pi x : a.b) &
\end{array}$$

(Def)

$$\begin{array}{lll}
\Delta & \Gamma & \varphi : a \\
\Delta & \mathbf{y} : \mathbf{B} & d(\mathbf{y}) : b \\
\Delta D & \Gamma & \varphi : a
\end{array}$$

where D is the string

$$\mathbf{y} : \mathbf{B} \quad c\mathbf{y} \equiv d(\mathbf{y}) : b$$

that $d(\mathbf{y})$ could be an empty word.

(Inst)

If Δ contains the element

$$\mathbf{y} : \mathbf{B} \quad c\mathbf{y} \equiv d(\mathbf{y}) : a$$

that $d(\mathbf{y})$ could be empty, and we have $\mathbf{U} \in \mathbf{F}^*$ with the same size of $\mathbf{y}$, then

$$\begin{array}{ll}
\Delta \quad \Gamma \quad * : \square & \\
\Delta \quad \Gamma \quad U_0 : B_0 \frac{U}{\mathbf{y}} & \\
\vdots & \\
\Delta \quad \Gamma \quad U_m : B_m \frac{U}{\mathbf{y}} & \\
\Delta \quad \Gamma \quad c\mathbf{U} : a \frac{U}{\mathbf{y}} &
\end{array}$$

Now we introduce the rule of inductions.

Suppose X is a constant symbol, $c_i \in \mathbf{C}_X$, and $\Pi(\mathbf{x} : \mathbf{A}).s \in \mathbf{A}$, then

$$\begin{array}{ll}
\Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s & c_0 : s \\
\vdots & \\
\Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s & c_m : s \\
\Delta \quad \Gamma \quad \text{Ind}X : \Pi(\mathbf{x} : \mathbf{A}).s &
\end{array}$$

and it immediately follows that

$$\begin{array}{ll}
\Delta \quad \Gamma \quad \text{Ind}X : \Pi(\mathbf{x} : \mathbf{A}).s & \\
\Delta \quad \Gamma \quad \text{Cstr}X c_i : c_i \frac{\text{Ind}X}{X} &
\end{array}$$

Assume that $X : \Pi(\mathbf{x} : \mathbf{A}).s$, $Y : \Pi(\mathbf{x} : \mathbf{A}).s'$ and $c \in \mathbf{C}_X$. This induces a function $E_{X,Y} : \mathbf{C}_X \rightarrow \mathbf{C}_Y$, which is defined inductively.

$$\begin{aligned}
E_{X,Y}(X) &= X \\
E_{X,Y}(cM) &= E_{X,Y}(c)M \\
E_{X,Y}(M \rightarrow c) &= M \rightarrow M[Y/X] \rightarrow E_{X,Y}(c) \\
E_{X,Y}(\Pi x : M.c) &= \Pi x : M.E_{X,Y}(c) \quad (X \notin M)
\end{aligned}$$

Lemma 1.1.6 *If for all s'' , $(s'', s) \in \mathcal{A}$ implies $(s'', s') \in \mathcal{A}$, and $(s', s') \in \mathcal{A}$, then the follows is derivable*

$$\begin{array}{ll}
\Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s & c : s \\
\Delta \quad \Gamma \quad \Pi(\mathbf{x} : \mathbf{A}).s' & \\
\Delta \quad \Gamma \quad X : \Pi(\mathbf{x} : \mathbf{A}).s & Y : \Pi(\mathbf{x} : \mathbf{A}).s' \quad E_{X,Y}(c) : s'
\end{array}$$

If $f_0, \dots, f_m$ and G are new constant symbols, we have

$$\begin{array}{ll}
\Delta \quad \Gamma \quad G : \text{Ind}X a & \\
\Delta \quad \Gamma \quad Y : \Pi(\mathbf{x} : \mathbf{A}).s' & \\
\Delta \quad \Gamma \quad f_i : E_{X,Y}(c_i) & \\
\Delta \quad \Gamma \quad \text{Elim}GY f_0 f_1 \dots f_m : Y a &
\end{array}$$